

AI-Based Feedback Systems in Piano Education: Effects on Technical Accuracy and Learner Autonomy

Sistemas de feedback baseados em IA no ensino do piano: efeitos na precisão técnica e na autonomia do aluno



Xiaodong Deng

Department of Keyboard Music Education, Shenyang Conservatory of Music, Shenyang, China

xiaodongdeng@chinesefineart.org

Abstract: This study aims to empirically evaluate whether using an AI-based feedback system, combined with traditional instruction, can improve technical accuracy and promote learner autonomy among piano students, compared with a control group receiving only conventional training. The sample consisted of 240 first-year piano majors. Learner autonomy was assessed using the Autonomous Educational Learning Scale, while technical accuracy in piano performance was evaluated through multimodal analysis employing Onsets & Frames and MediaPipe software. The findings confirmed the positive impact of the AI-based feedback system (PFS-AI, Onsets & Frames, MediaPipe) on both learner autonomy and technical proficiency. A mixed-design ANOVA revealed significant main effects of time and group, as well as their interaction. The study offers practical and scholarly insights by empirically validating the effectiveness of AI-driven feedback systems, suggesting their potential to reduce instructional burden and enhance the quality of piano education.

Keywords: artificial intelligence; feedback systems; music pedagogy; musical performance quality; piano practice analysis.

Resumo: Este estudo teve como objetivo avaliar empiricamente se a utilização de um sistema de feedback baseado em IA, em combinação com o ensino tradicional, pode melhorar a precisão técnica e promover a autonomia do aluno de piano, em comparação com um grupo de controle que recebeu apenas treino convencional. A amostra foi composta por 240 alunos do primeiro ano de piano. A autonomia do aluno foi avaliada por meio da Escala de Aprendizagem Educativa Autônoma (AELS), enquanto a precisão técnica na execução pianística foi avaliada por meio de análise multimodal, com o uso dos softwares Onsets & Frames e MediaPipe. Os resultados confirmaram o impacto positivo do sistema de feedback baseado em IA (PFS-AI, Onsets & Frames, MediaPipe) tanto na autonomia do aluno quanto na proficiência técnica. O método estatístico ANOVA de delineamento misto revelou efeitos principais significativos do tempo e do grupo, bem como a interação entre eles. O estudo oferece insights práticos e acadêmicos ao validar empiricamente a eficácia dos sistemas de feedback baseados em IA, sugerindo seu potencial para reduzir a carga de trabalho dos professores e melhorar a qualidade do ensino de piano.

Palavras-chave: Inteligência artificial; sistemas de feedback; pedagogia da performance; qualidade da performance musical; análise da prática de piano..

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1. Introduction

In recent years, artificial intelligence (AI) technologies have advanced rapidly, and in the context of music education, an increasing number of studies have focused on developing AI-based feedback systems for performance analysis (Cooper, 2026; Dewanto and Sugiharto, 2025; Wang et al., 2024). These systems employ machine learning algorithms, audio signal processing, MIDI data analysis, and computer vision, enabling them to replicate a core function of music instructors – monitoring and correcting performance (Gu, 2025). Within this paradigm, AI emerges as a crucial component of the educational process, with the potential to transform the acquisition of piano technique (Bai, 2021; Loo et al., 2021). However, the successful integration of AI technologies requires a clear understanding of which aspects of performance can be assessed objectively and how such innovations may influence learning processes (Laak and Aru, 2025). For example, technical accuracy in piano performance, traditionally evaluated subjectively by instructors – often resulting in variability of feedback – can be standardized using digital metrics (Evin, 2024). At the same time, piano instruction extends beyond mere technical accuracy, encompassing such dimensions as interpretation, expressiveness, and communicative performance. In professional music pedagogy, technical skills are generally viewed not as ends in themselves but as instruments that enable the stylistic interpretation of a musical work and the realization of its expressive potential, thereby establishing a connection between the performer and the audience (Jia, 2023; Moura et al., 2024). Although artificial intelligence-based feedback systems demonstrate considerable effectiveness in analyzing objective technical parameters, including tempo stability, rhythmic consistency, and note accuracy (Chang and Peng, 2022; Evin, 2024; Zhang, 2024), their capacity to evaluate the subtleties of expressiveness, artistic interpretation, and emotional communication remains limited (Lerch et al., 2020; Liao, 2025). Recognition of these

limitations is important for the appropriate understanding and integration of AI-based feedback within the broader context of contemporary music pedagogy and the performing arts. This new AI form of feedback also intersects with another essential component of music education: learner autonomy, which encompasses the ability to self-analyze performance, address errors, and independently improve (Rodriguez, 2025). AI systems are expected to strengthen these skills by offering personalized recommendations and continuous access to performance analysis. However, it remains unclear whether such technologies genuinely foster autonomy or inadvertently create overreliance on external tools (Chang and Tsai, 2024; Huang, 2024; Li, 2022).

Exploring how interaction with AI influences the development of self-regulatory skills has thus become a critical area of inquiry. One promising approach is multimodal assessment, a comprehensive methodology that integrates diverse data sources – audio recordings, behavioral interaction metrics, and MIDI files – providing a more accurate evaluation of technical performance and valuable insights into students' engagement with AI systems (Chang and Peng, 2022; Zhang, 2024). Despite the growing availability of AI-based tools, current research in this field remains fragmented both globally and within the Chinese context (Bai, 2021; Su et al., 2025). Many existing studies are limited by small sample sizes or focus solely on the theoretical potential of such technologies without empirical validation (Dewanto and Sugiharto, 2025; Evin, 2024; Zhang, 2024). Furthermore, the use of multimodal indicators in the literature is still insufficiently widespread, and real-world student interaction with AI-based feedback systems remains largely underexplored (Chang and Peng, 2022; Lee and So, 2025; Li, 2022). These gaps highlight ongoing uncertainties regarding the effectiveness of digital feedback systems and underscore the need for further empirical investigation, thereby defining the relevance of the present study.

The study seeks to empirically assess whether using an AI-based feedback system, combined with traditional instruction, can enhance technical accuracy and foster learner autonomy among piano students, compared with a control group receiving only conventional training. It also aims to analyze the nature of the experimental group's interaction with the AI system. Synthesizing all collected data may help clarify the mechanisms underlying the effectiveness of AI feedback and its impact on the development of both autonomy and technical skills in piano education. The scholarly significance of this article lies in the application of a multimodal methodology that incorporates interaction metrics between students and the AI system. The research addresses a gap in the literature: the lack of controlled experimental studies comparing traditional instruction with AI-enhanced approaches. The practical value of the study is reflected in its potential to inform the optimization of digital learning platforms, support the development of adaptive training systems, and improve the effectiveness of independent student practice. The findings may be particularly relevant for music institutions seeking to implement AI technologies in their educational practices.

1.1 Literature Review

According to recent studies, the literature presents a wide range of applications and tools designed to provide individualized recommendations and real-time performance evaluation (Evin, 2024; Lu, 2025; Ou et al., 2025). These systems are based on audio signal processing, pattern recognition, and machine learning algorithms, enabling the analysis of students' performance and delivering immediate feedback on accuracy, rhythmic stability, and playing quality (Li, 2025). Within this paradigm, AI serves as a digital tutor that supports independent learning, partially complementing or replacing the instructor (Xu and Fan, 2024).

Particular attention has been given to the use of AI in piano education. Contemporary platforms integrate with digital keyboards or microphones and can automatically capture played notes, track their dynamic execution, compare them to a reference score, and visually highlight errors (Dewanto and Sugiharto, 2025). Feedback is delivered by identifying incorrect notes, signaling rhythmic accuracy, and digitally scoring each attempt, thus ensuring regular and objective assessment of technical accuracy without instructor intervention (Wang, 2025). Empirical studies confirm the importance of timely and precise feedback in instrumental training, particularly in early stages when correcting errors has a significant impact on learners' progress (Borja and Mendoza, 2024; Ou et al., 2025). Research highlights the successful use of automatic music transcription methods powered by deep learning technologies, which enable alignment of played melodies with the musical score and help reduce the risk of reinforcing incorrect motor patterns (Ru, 2025; Tang, 2025).

When comparing traditional piano lessons with the integration of AI-based platforms, participants often show improvement in key learning outcomes, although short-term gains in technical skills do not always exceed those achieved through conventional instruction (Yeh et al., 2025). This underlines the importance of combining automated tools with pedagogical mentorship.

Literature on autonomy and self-regulation in music education emphasizes the critical role of learner independence in mastering complex skills (Oguguo et al., 2022). Autonomy is defined as the learner's willingness and capacity to take responsibility for their own learning outcomes (Holec, 1981) and is closely associated with self-regulated learning, which is particularly relevant in the context of musical practice (Admiraal et al., 2024). Studies in music pedagogy demonstrate that self-efficacy and self-regulation skills are reliable predictors of performance success (Can, 2025; Zelenak, 2024).

In this context, AI-based feedback systems receive mixed evaluations. On one hand, such technologies provide opportunities for more independent learning by offering resources that allow students to practice without continuous teacher involvement (Li, 2025). On the other hand, there is a risk that students may relinquish their active role in the learning process and rely entirely on algorithms for analysis and decision-making (Hyde et al., 2024). This raises the imperative to design didactically balanced AI tools that do not replace student initiative but rather support and enhance it.

Regarding technical accuracy, its core components typically include note correctness, rhythmic organization, sound clarity, and articulation precision (Lerch et al., 2020). Analyses of performance assessment criteria in piano playing indicate that instructors often distinguish between technical and interpretative dimensions of execution (Moura et al., 2024; Russell, 2014). Technical accuracy is generally regarded as a foundational prerequisite for developing artistic expressiveness and emotional depth that transcend mechanical reproduction (Jia, 2023). In this context, there is growing interest in objective, automated evaluation methods, including multimodal analysis that integrates audio and video data to assess performance from multiple perspectives (Li, 2020).

Contemporary algorithms can extract a wide range of characteristics from audio signals, including pitch, duration, and loudness, as well as spectral and timbral features (Christodoulou et al., 2024). Nevertheless, some studies indicate that capturing subtle nuances of dynamics and timbre remains a challenge, requiring further refinement of spectral analysis techniques (Liao, 2025; Wei, 2021). At the same time, existing models can classify piano performances based on combinations of pitch and timbral parameters (Zhu, 2024). This enables quantitative assessment of aspects such as error rates, rhythmic deviations, and dynamic stability, and supports objective comparisons of student progress across groups (Chang and Peng, 2022; Zhang,

2024). Although the literature emphasizes the importance of autonomy and self-regulatory skills in music education (Admiraal et al., 2024; Can, 2025; Oguguo et al., 2022; Zelenak, 2024), empirical evidence demonstrating that regular use of digital feedback enhances independence rather than fostering reliance on algorithmic suggestions remains scarce (Hyde et al., 2024). Furthermore, current research seldom addresses the nature of learners' interaction with AI systems, focusing instead on final performance outcomes (Dai et al., 2025; Ou et al., 2025; Yeh et al., 2025). Another concern is the limited scope of assessment metrics used in existing studies, which tend to prioritize audio-based indicators and rarely incorporate multimodal evaluation (Christodoulou et al., 2024).

Nevertheless, multimodal approaches hold significant potential for increasing the objectivity and accuracy of performance analysis (Lerch et al., 2020; Moura et al., 2024; Russell, 2014). Additionally, there is a notable lack of studies comparing multimodal AI systems with traditional in-person instruction under controlled experimental conditions (Borja and Mendoza, 2024; Li, 2025; Xu and Fan, 2024). This limitation hinders a comprehensive assessment of the actual advantages offered by such technologies. These research gaps highlight the need for systematic empirical investigations in this domain.

1.2 Theoretical Framework

In music pedagogy, technical accuracy is perceived not merely as an independent mechanical skill, but as a fundamental prerequisite for achieving expressive freedom, stylistic interpretation, and artistic communication (Jia, 2023; Moura et al., 2024; Russell, 2014). From this perspective, the development of piano performance technique is closely associated with the process of interpretative decision-making and the realization of musical expression. Accordingly, artificial intelligence-based feedback systems should be regarded primarily as tools that facilitate the objective improvement of technical aspects, rather

than as substitutes for the interpretative and communicative processes traditionally cultivated through teacher–student interaction. An analysis of the present study through the lens of this broader pedagogical perspective yields a more comprehensive understanding of the educational value of AI-based feedback and highlights the existing limitations of AI applications in the teaching of the performing arts.

This study is grounded in the theory of self-regulated learning (Zimmerman, 2002), which posits that skill development occurs over time through practice, feedback, and reflection. Drawing on principles from feedback psychology (Hattie and Timperley, 2007) and self-determination theory (Ryan and Deci, 2000), high-quality feedback can be assumed to effectively satisfy learners' cognitive need for competence. These theoretical premises inform Hypothesis 1 (H1), which proposes that the use of a system providing immediate, objective, and bias-free feedback promotes more deliberate management of the learning process by students, thereby resulting in higher levels of learner autonomy.

Furthermore, the concept of human–AI cognitive partnership in educational environments (Algammaz, 2025; Cukurova et al., 2019; Goldie, 2016) positions technology as a learning partner capable of delivering personalized recommendations. This perspective allows instructional resources to be reallocated toward addressing more complex and nuanced aspects of content mastery. It is anticipated that continuous monitoring and error correction through multimodal assessment of technical performance will accelerate skill acquisition. Based on this framework, Hypothesis 2 (H2) asserts that integrating AI-generated feedback will lead to greater improvements in technical accuracy among students in the experimental group than among those trained solely under the guidance of a human instructor. This hypothesis is supported by findings from a recent study involving violinists (Ou et al., 2025), which reported enhanced performance quality following the implementation of an AI application.

Finally, Hypothesis 3 (H3) stems from the assumed benefits of AI-based feedback systems and posits that students who engage more actively with the system's features will exhibit more significant gains in both technical proficiency and learner autonomy.

1.3 Problem Statement

The primary aim of this study is to empirically determine whether an AI-based feedback system, when combined with traditional instruction, can foster learner autonomy and enhance technical accuracy in piano performance compared with traditional instruction alone. In addition, the study seeks to gather data on students' interactions with the AI feedback system to explore the extent of its use and its relationship to both autonomy and the acquisition of technical skills.

To achieve these objectives and address the three proposed hypotheses, the following research tasks were formulated:

(1) To assess the impact of the AI-based feedback system on the development of student autonomy compared to traditional instruction alone.

(2) To compare the dynamics of technical accuracy development in piano performance between the control group (traditional instruction only) and the experimental group (traditional instruction combined with AI-based feedback).

(3) To examine whether the intensity of interaction with the AI-based feedback system among students in the experimental group is associated with improvements in technical accuracy and learner autonomy, and to explore the nature of this relationship.

2. Methods and Materials

In the present study, learner autonomy was assessed using the Autonomous Educational Learning Scale (AELS), which has been validated for use in China (Lin and Reinders, 2017). The instrument consists of 32 items specifically adapted to reflect autonomy in piano learning and is structured into three subscales: (1) self-management skills, which include goal setting, planning, resource selection, self-monitoring, and the ability to adjust learning strategies; (2) autonomous learning behavior, which captures the frequency and quality of independent practice, initiative in task completion, and use of supplementary materials; and (3) autonomous psychology, which refers to the learner's willingness to take responsibility for outcomes, confidence in their abilities, awareness, and intrinsic motivation. All items are rated on a 5-point Likert scale, with higher scores indicating greater autonomy. Scores are calculated separately for each subscale, and an overall autonomy index is then derived. The possible score range is 32 to 160.

Technical accuracy was examined through multimodal analysis based on audio, MIDI, and video data collected during student performances of three piano segments, every 16 measures in length (approximately 35–40 seconds of playing). For audio analysis, the Onsets & Frames neural network model (Cheuk et al., 2021) was used to generate high-precision MIDI data, capturing the played notes, onset times, durations, and dynamics. This data enabled the calculation of three performance metrics as percentages: (1) correct notes, (2) rhythmic deviations, and (3) tempo stability.

Video analysis was conducted using the computer vision library MediaPipe (Lugaresi et al., 2019), which automatically detects the human skeletal structure. In this study, three additional indicators of technical accuracy were derived from video data, also expressed as percentages: (1) accuracy of

hand movements (including fingers and wrists), (2) movement stability within an optimal amplitude range, and (3) precision of key presses relative to the reference rhythm (timing accuracy). All six indicators – three from Onsets & Frames and three from MediaPipe – were automatically computed and expressed as percentages, providing quantitative data for performance evaluation.

2.1 Instrument Validation

Although the AELS scale was originally developed and validated in Chinese for use in China, it underwent a standardized evaluation procedure to confirm its suitability for the current research design. During the pilot testing phase ($n = 100$), exploratory factor analysis (EFA with Varimax rotation) confirmed the three-factor structure of the scale, with factor loadings ranging from 0.56 to 0.82. Confirmatory factor analysis (CFA) indicated satisfactory model fit indices ($CFI = 0.946$, $TLI = 0.951$, $RMSEA = 0.049$). Reliability estimates for the AELS subscales were as follows: self-management skills ($\alpha = 0.85$), autonomous behavior ($\alpha = 0.84$), autonomous psychology ($\alpha = 0.88$), and the overall autonomy index ($\alpha = 0.86$). Composite reliability (CR) ranged from 0.84 to 0.92, McDonald's omega (Ω) from 0.82 to 0.94, and the average variance extracted (AVE) from 0.52 to 0.63, all meeting the recommended threshold of > 0.50 .

The technical accuracy indicators derived from Onsets & Frames and MediaPipe, expressed as percentages, demonstrated intermodal consistency ($r = 0.71$ – 0.79) and high test-retest reliability, as indicated by intraclass correlation coefficients ($ICC = 0.85$ – 0.92).

2.2 Participants

The study involved 240 first-year students from the Shenyang Conservatory of Music. Recruitment was conducted through campus announcements, email distributions, and

in-class invitations. Inclusion criteria required participants to be at least 18 years old, enrolled in a piano-focused music degree program in their first year, able to participate in multimodal performance recordings, and free from conditions that might interfere with performance (e.g., physical injury). Students were excluded if they were non-Chinese nationals, had diagnosed auditory, motor, or cognitive impairments, or were already using AI-assisted music practice systems at the start of the study. Detailed demographic and background information is provided in Table 1.

Table 1. Data on the participants of the current study

Group	Total	Male	Female	Mean Age	SD
Control	120	59	61	18.51	0.24
Intervention	120	58	62	18.31	0.29
Mean/Total	240	117	123	18.41	0.26

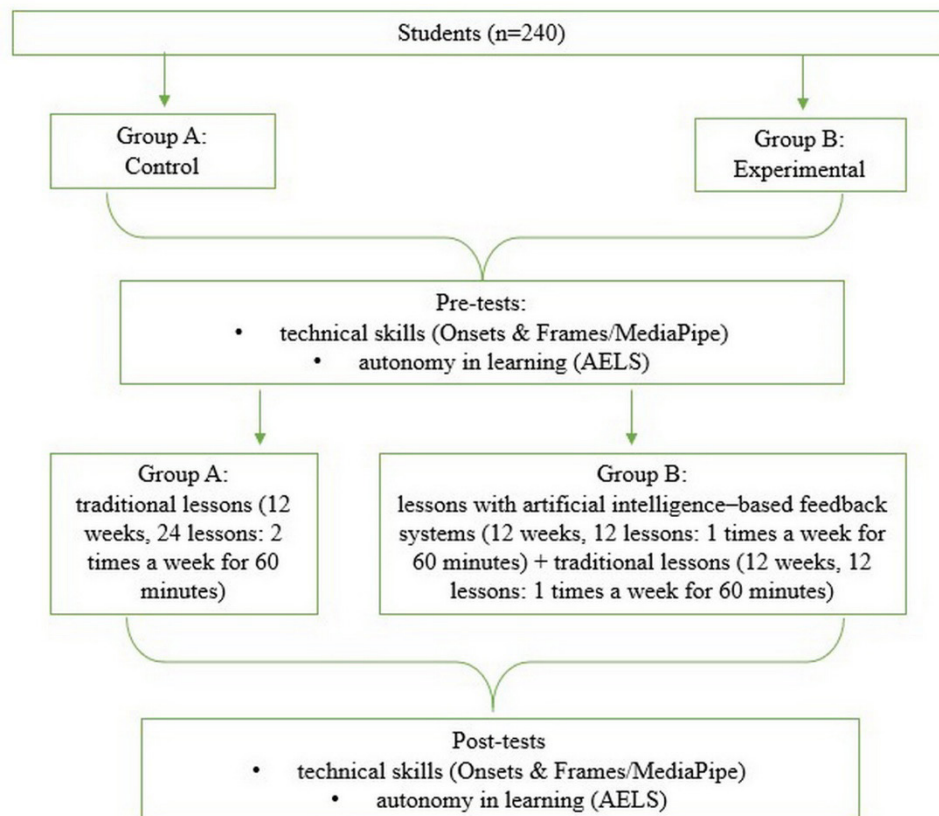
At the pre-test stage, candidates were stratified by their level of piano performance proficiency – categorized as beginner, intermediate, or advanced – based on prior experience, practice history, and individual performance indicators obtained from Onsets & Frames and MediaPipe data. This stratification ensured a balanced distribution of technical skill across groups. As a result, 120 students were assigned to the control group, which received traditional instruction without digital technologies, while the remaining 120 students were assigned to the experimental group, which received piano instruction incorporating AI-based feedback systems.

The sample included 117 male participants (48.8%) and 123 female participants (51.2%), with a mean age of 18.41 years (SD = 0.26). Following the group assignment, a baseline equivalence check was conducted. Statistical analysis revealed no significant differences between the control and experimental groups in either technical performance or autonomy levels ($p > 0.05$), confirming the equivalence of the groups at the starting point of the study.

2.3 Study Design

The design scheme of the present study is illustrated in Figure 1.

Figure 1. Overview of the study design



The study was initiated in February 2025, at which point participants were recruited and baseline assessments were conducted. These included pre-tests of technical performance using Onsets & Frames and MediaPipe, as well as pre-tests of learning autonomy using the AELS scale. Following these assessments, participants were stratified and assigned to either the control or experimental group. Group equivalence was verified in terms of both technical accuracy and autonomy before the intervention began.

The control group continued with their standard instructional format, receiving two 60-minute piano lessons per week conducted in person with a teacher. These traditional lessons focused on repertoire assigned by the curriculum, technical exercises (scales, arpeggios, etudes), and refinement of musical elements such as dynamics, rhythm, and articulation. Teachers provided real-time correction and individualized feedback.

In contrast, the experimental group received one traditional 60-minute piano lesson per week, and an additional 60-minute session each week as part of the intervention program. During the intervention session, students practiced compositions selected by the AI program and received automated feedback from the system. The intervention lasted for 12 weeks. At the end of May 2025, all participants completed post-tests of technical performance (Onsets & Frames and MediaPipe) and learner autonomy (AELS). All data were collected in digital format and used for subsequent analysis.

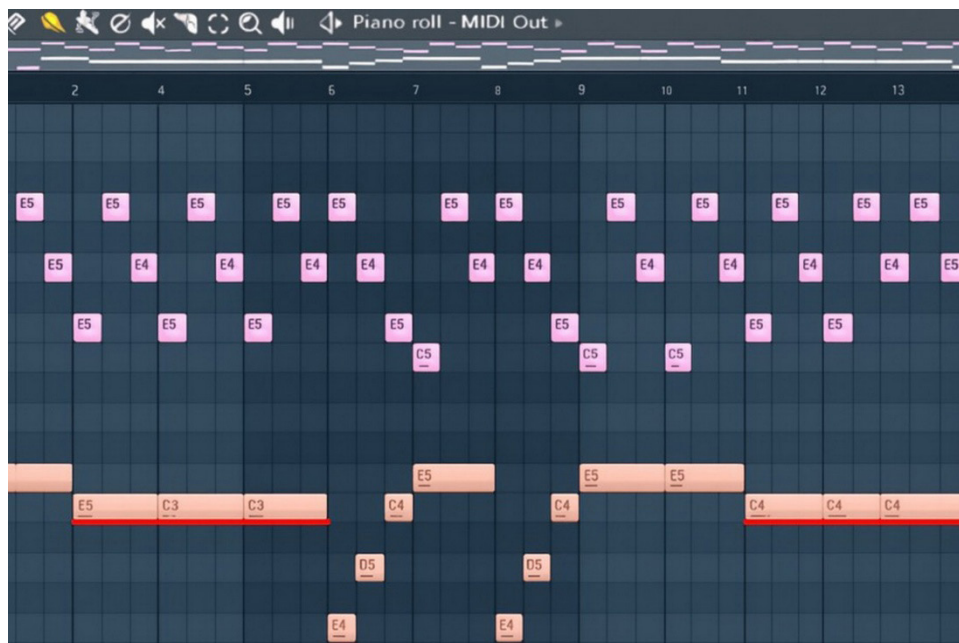
2.4 Intervention

The AI-based feedback system used in the intervention was the Piano Feedback System-AI (PFS-AI), specialized software designed to automatically assess piano performance using artificial intelligence. This system analyzes unstructured musical input and transforms it into structured indicators of performance quality. It has previously been implemented in other cultural contexts (Algammaz, 2025).

PFS-AI operates by processing a large volume of student-generated MIDI and audio recordings. The system is pre-trained on thousands of performance variations of the same composition, enabling it to distinguish between correct and incorrect notes, assess rhythmic stability, evaluate legato

and staccato quality, phrasing accuracy, tempo consistency, bar alignment, beat placement, and other performance elements. The PFS-AI system was trained on an extensive dataset comprising annotated piano performances, including diverse examples of educational repertoire performed by both students and professional musicians. The primary focus of the system's evaluation was on practically measurable technical parameters, including note accuracy, rhythmic precision, tempo stability, articulation clarity, and motor coordination. However, the system was not originally designed for the independent assessment of broader interpretative dimensions, such as artistic originality, emotional expressiveness, stylistic nuance, or the ability to engage with an audience. The feedback provided by the system was therefore based on the algorithmic evaluation of predefined technical characteristics rather than on a comprehensive artistic analysis. Each note, measure, and expressive gesture is automatically classified by the system and compared to a reference standard. Upon completion of the performance, PFS-AI generates an objective feedback report displayed on the laptop screen that includes both audio and graphical feedback (Figure 2), reinforcing students' efforts through positive messaging while constructively identifying technical errors in a pedagogically sensitive manner.

Figure 2. Example of PFS-AI feedback displayed on screen, indicating user errors (highlighted in red)



In addition, the PFS-AI system was enhanced with computer vision and machine learning components (Yang, 2025). Specifically, the following tools were integrated: (1) the OpenPose neural network for skeletal tracking, which automatically extracted the coordinates of 32 anatomical keypoints from the video stream; (2) dynamic time warping (DTW), which was used to analyze hand and upper-body movement patterns during piano playing by aligning individual motion trajectories with previously recorded reference movements and identifying deviations in technique through DTW-based evaluation; (3) the k-nearest neighbors (K-NN) method for classifying posture based on a feature vector derived from joint coordinates, enabling the system to automatically detect postural states such as “correct posture,” “forward lean,” “slouching,” and other common errors in instrumental performance.

Figures 3 and 4 present screenshots of video-based recordings generated using OpenPose, DTW, and K-NN, illustrating skeletal and hand tracking during piano performance.

Figure 3. Screenshots of video-based recordings generated by OpenPose, DTW, and K-NN for skeletal tracking during piano performance

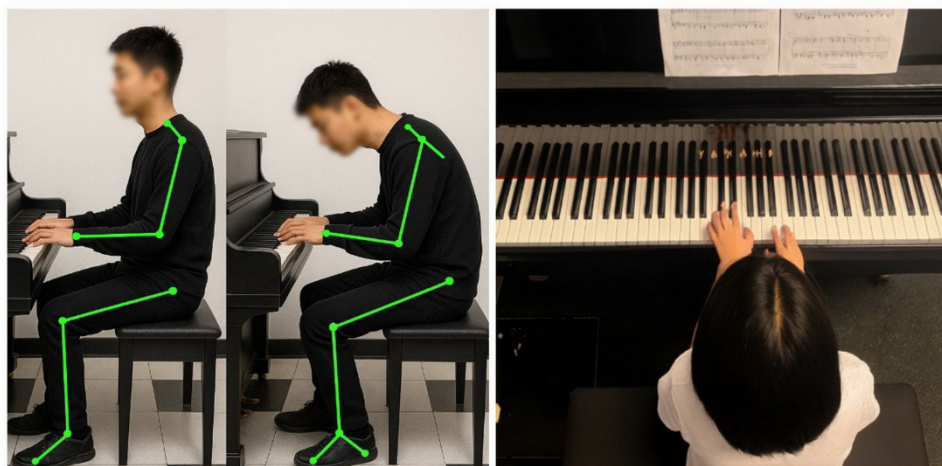
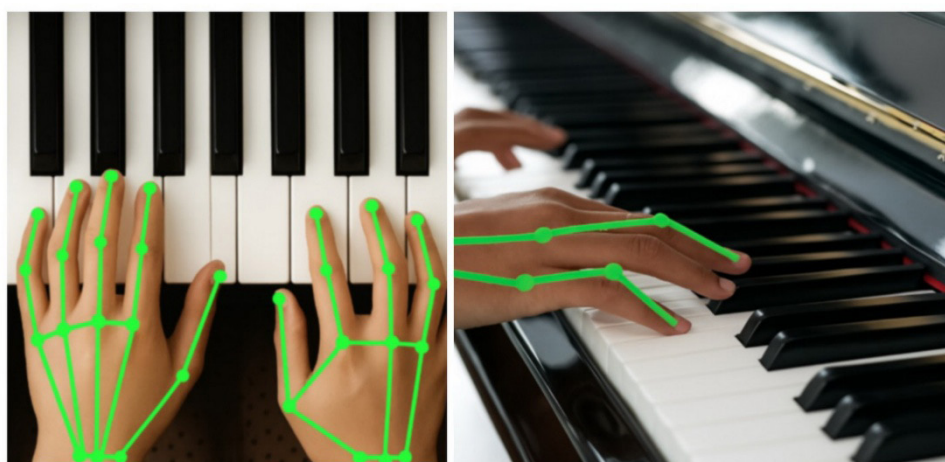


Figure 4. Screenshot of video-based recording using OpenPose, DTW, and K-NN for hand tracking during piano performance



The system operates by capturing the musician's movements in real time using a convolutional neural network (CNN), while dynamic time warping (DTW) analyzes deviations in movement patterns, and the k-nearest neighbors (K-NN) algorithm classifies the musician's posture. Based on the combined input from these sources, the system immediately generates auditory prompts when an error is detected, creating a continuous loop of corrective feedback.

To ensure accurate functioning of the AI system, six stationary HD cameras were used to record the student's posture and full-body position, including hand movements and foot placement on the pedals. Audio was captured using an external condenser microphone mounted above the instrument, 40–50 cm from the soundboard, to ensure high-quality recording. The computer running the integrated PFS-AI system, along with OpenPose, DTW, and K-NN, was positioned on a separate table to the right of the instrument. A visual feedback display showing analytical output was placed directly in front of the student at eye level (above the sheet music stand) to minimize distraction during performance. All lessons took place in specially equipped classrooms on campus.

2.5 Data Analysis

All statistical analyses were conducted using IBM SPSS Statistics 27. To verify baseline group equivalence in technical skills and autonomy, independent samples t-tests were performed. Descriptive statistics, including means and standard deviations, were calculated, along with paired-sample nonparametric t-tests. Intervention effects were evaluated using partial eta-squared (η^2p). Statistical significance was set at $p < 0.05$, and Holm–Bonferroni correction was applied for multiple comparisons.

To assess the progression of the target outcomes (technical accuracy and autonomy), a mixed-design ANOVA was conducted, with time (pre-test vs. post-test) as the within-subject factor and group (control vs. experimental) as the between-subject factor, along with their interaction. The relationship between the intensity of interaction with the AI system and final outcomes in technical accuracy and autonomy was analyzed using multiple regression. Predictor variables included system log data (number of sessions, average session duration, proportion of accepted system recommendations, and implementation success rate), which were extracted directly from Onsets &

Frames and MediaPipe. Dependent variables were the delta values (gains) in technical and autonomy scores within the experimental group.

2.6 Ethical Issues

The study adhered to the ethical principles outlined in the Declaration of Helsinki. Ethical approval for the project was granted by the Ethics Committee of the Shenyang Conservatory of Music. All participants were adults and voluntarily agreed to take part in the study. They were assured of the anonymity and confidentiality of their personal data. All participant data collected during the study were stored in a de-identified (coded) format on a secure server and were accessible only to the research team.

3. Results

Table 2 presents descriptive statistics (means and standard deviations) for the pre-test and post-test scores on learner autonomy (AELS) and the six technical accuracy indicators derived from Onsets & Frames and MediaPipe. Table 3 displays the results of the paired-sample nonparametric t-test used to compare pre- and post-test scores within each group to assess statistical significance.

Table 2. Descriptive statistics (M and SD) for pre-test and post-test scores on autonomy (AELS) and six technical accuracy indicators obtained using Onsets & Frames and MediaPipe

Indicator	Group	M	SD
AELS (pre-test)	Control	70.10	1.393
	Intervention	70.05	1.263
AELS (post-test)	Control	70.42	1.089
	Intervention	76.20	3.268
Correct notes (pre-test)	Control	69.93	3.264
	Intervention	70.03	3.430

Correct notes (post-test)	Control	77.19	1.702
	Intervention	81.18	2.007
Rhythmic deviations (pre-test)	Control	71.63	4.442
	Intervention	71.63	4.491
Rhythmic deviations (post-test)	Control	78.45	1.767
	Intervention	77.78	2.225
Tempo stability (pre-test)	Control	69.32	2.394
	Intervention	68.77	2.485
Tempo stability (post-test)	Control	76.42	1.127
	Intervention	80.80	1.459
Hand movements (pre-test)	Control	68.80	3.756
	Intervention	69.63	3.719
Hand movements (post-test)	Control	77.42	1.781
	Intervention	80.66	1.836
Movement stability (pre-test)	Control	71.13	5.664
	Intervention	70.66	5.212
Movement stability (post-test)	Control	78.36	1.664
	Intervention	79.88	2.087
Key press accuracy (pre-test)	Control	67.41	1.808
	Intervention	67.33	1.755
Key press accuracy (post-test)	Control	76.57	1.090
	Intervention	81.25	1.404

Table 3. Paired nonparametric t-test results for pre-test and post-test scores on autonomy (AELS) and six technical accuracy indicators

Indicator		Mean difference	t	df	p
Control	AELS (autonomy)	-0.317	-1.989	119	0.049
	Correct notes	-7.258	-20.160	119	< 0.001
	Rhythmic deviations	-6.825	-14.721	119	< 0.001
	Tempo stability	-7.100	-28.459	119	< 0.001
	Hand movements	-8.625	-22.653	119	< 0.001
	Movement stability	-7.233	-13.617	119	< 0.001
	Key press accuracy	-9.167	-44.845	119	< 0.001

Intervention	AELS (autonomy)	-6.150	-19.494	119	< 0.001
	Correct notes	-11.142	-33.300	119	< 0.001
	Rhythmic deviations	-6.142	-13.928	119	< 0.001
	Tempo stability	-12.025	-47.822	119	< 0.001
	Hand movements	-11.033	-27.834	119	< 0.001
	Movement stability	-9.225	-17.411	119	< 0.001
	Key press accuracy	-13.917	-67.160	119	< 0.001
Note: significant at $p < 0.05$					

Autonomy scores and all six technical accuracy indicators (Table 2) increased in both the control and experimental groups. However, the gains were significantly higher across all variables in the experimental group. Specifically, autonomy in the experimental group increased by 6.15 points ($p < 0.001$), while the control group showed only a marginal increase of 0.32 points, which was not statistically significant ($p = 0.49$). In terms of technical accuracy, the experimental group improved by 11 to 14 percentage points, as reflected in high t -values (e.g., $t = -47.822$ for tempo stability and $t = -67.160$ for key press accuracy), whereas the t -values for the control group were notably lower.

To evaluate the effect of the AI-based feedback system on the development of learner autonomy compared with traditional instruction alone, a mixed-design analysis of variance was conducted, incorporating time and group as factors, as well as their interaction. The corresponding data are presented in Table 4.

Table 4. Mixed ANOVA results for autonomy (AELS)

Indicator	Factor	F	p	η^2p
AELS	Time	148.62	< 0.001	0.384
	Group	96.41	< 0.001	0.287
	Group $\times$ Time	182.77	< 0.001	0.433
Note: significant at $p < 0.05$				

The results revealed a significant main effect of time ($p < 0.001$, $\eta^2p = 0.384$), indicating overall changes in learner autonomy across the sample. A significant between-group effect

($p < 0.001$, $\eta^2p = 0.287$) suggests differences in autonomy levels between the control and experimental groups. Additionally, the group $\times$ time interaction effect was significant ($p < 0.001$, $\eta^2p = 0.433$), indicating divergent trajectories of autonomy development over time between the two groups.

To compare the progression of technical accuracy in piano performance between the control and experimental groups, mixed-design ANOVA values were calculated (Table 5).

Table 5. Mixed ANOVA results for six technical accuracy indicators obtained using Onsets & Frames and MediaPipe

Indicator	Factor	F	p	η^2p
Correct notes	Time	812.54	<0.001	0.774
	Group	128.91	<0.001	0.351
	Group $\times$ Time	256.37	<0.001	0.518
Rhythmic deviations	Time	426.12	<0.001	0.642
	Group	4.38	0.038	0.018
	Group $\times$ Time	96.44	<0.001	0.287
Tempo stability	Time	913.87	<0.001	0.792
	Group	189.66	<0.001	0.443
	Group $\times$ Time	318.52	<0.001	0.571
Hand movements	Time	690.55	<0.001	0.744
	Group	102.79	<0.001	0.302
	Group $\times$ Time	141.12	<0.001	0.372
Movement stability	Time	364.21	<0.001	0.605
	Group	19.55	<0.001	0.076
	Group $\times$ Time	55.33	<0.001	0.188
Key press accuracy	Time	1720.44	<0.001	0.877
	Group	310.52	<0.001	0.566
	Group $\times$ Time	612.78	<0.001	0.720
Note: significant at $p < 0.05$				

The results demonstrate a consistent and statistically significant improvement in technical accuracy across all six indicators, as evidenced by large time effects ($p < 0.001$, $\eta^2p = 0.605$ – 0.877). The main effects of the group indicate that

students in the experimental group exhibited higher levels of technical accuracy by the end of the intervention, although statistically significant improvements were also observed in the control group. The group $\times$ time interaction effect was significant for all indicators, with greater gains in technical skills observed in the experimental group (e.g., $\eta^2p = 0.518$ for correct notes and $\eta^2p = 0.720$ for key-press accuracy).

To determine whether and how the intensity of interaction with the AI-based feedback system was associated with improvements in technical accuracy and learner autonomy in the experimental group, a multiple regression analysis was conducted. The results are presented in Table 6 and illustrated in Figures 5 and 6.

Table 6. Multiple regression analysis results for predictors of interaction with the AI-based feedback system in relation to delta (gain) scores in technical accuracy and autonomy within the experimental group

Predictor	B	Std. Error	t	Sig.	95% CI	
					(Lower)	(Upper)
Δ Autonomy						
Number of sessions	0.07	0.03	2.07	0.056	0.02	0.14
Average session duration	0.03	0.02	1.52	0.131	−0.01	0.07
Proportion of accepted system recommendations	2.10	0.85	2.47	0.015	0.41	3.79
Success rate of recommendation implementation	2.85	0.74	3.86	<0.001	1.39	4.31
Δ Technical skills						
Number of sessions	0.35	0.07	4.86	<0.001	0.21	0.49
Average session duration	0.09	0.05	1.80	0.074	−0.01	0.19
Proportion of accepted system recommendations	4.25	1.31	3.24	0.002	1.66	6.84
Success rate of recommendation implementation	5.90	1.12	5.27	<0.001	3.69	8.11
* Note: significant at p-value 0.05						

Figure 5. Multiple linear regression illustrating the relationship between interaction intensity with the AI-based feedback system and the increase in learner autonomy

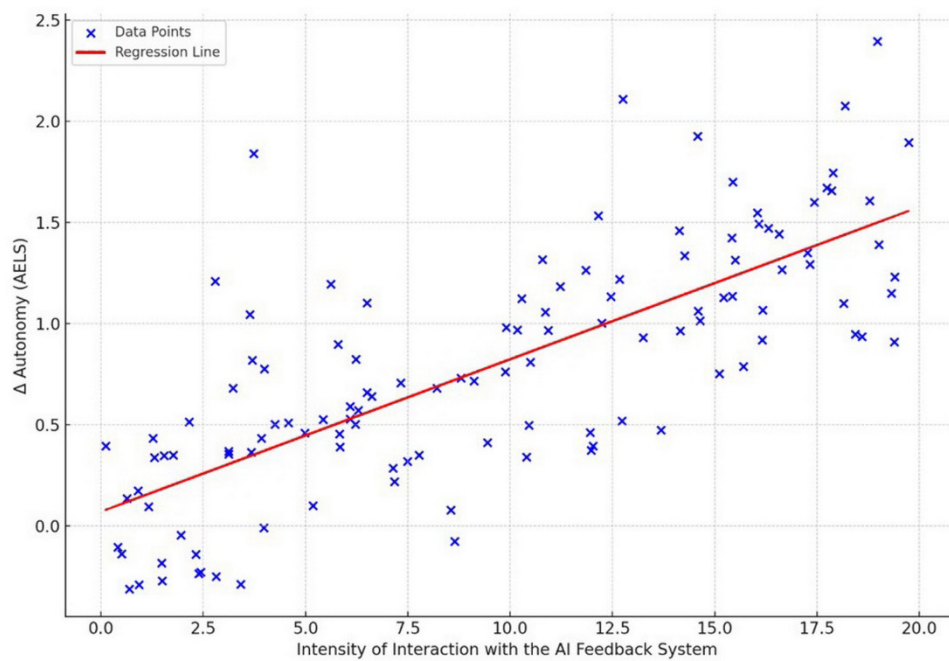
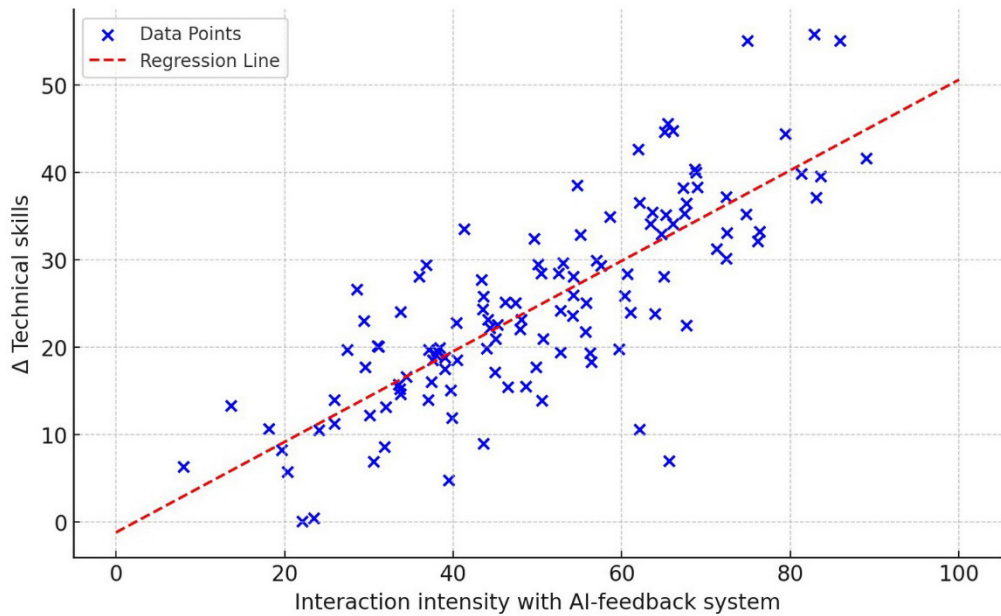


Figure 6. Multiple linear regression illustrating the relationship between interaction intensity with the AI-based feedback system and the increase in technical accuracy



The results indicate that the intensity of interaction with the AI-based feedback system was a significant predictor of both increased learner autonomy and improved technical skills. For autonomy, the strongest predictors were the success rate of recommendation implementation ($B = 2.85, p < 0.001$) and the proportion of accepted system recommendations ($B = 2.10, p = 0.015$), suggesting that the quality of engagement with the system plays a critical role in fostering learner independence.

Similarly, for technical skills, the most influential predictors were also the success rate of recommendation implementation ($B = 5.90, p < 0.001$) and the proportion of accepted recommendations ($B = 4.25, p = 0.002$), indicating a strong link between active correction of performance and the development of execution skills. The number of sessions also proved significant ($B = 0.35, p < 0.001$), further confirming that more frequent use of the system is associated with greater progress in piano performance.

Figure 5 visualizes a moderate positive association, although the data show substantial variability. In contrast, Figure 6 shows a stronger, more stable linear relationship, suggesting that interaction intensity with the AI system more accurately predicts improvements in technical accuracy.

4. Discussion

The results demonstrated a clear advantage of the AI-based feedback system in fostering both student autonomy and technical accuracy. While progress in these outcomes was observed in both the control and experimental groups, gains in the experimental group were significantly higher. The interaction effects in the mixed-design ANOVA were large (η^2p ranging from 0.188 to 0.720), indicating that skill development was accelerated in the context of AI-enhanced feedback. Regression analysis showed that the most significant predictors of improvements in both autonomy and technical skills were

qualitative indicators of system engagement, specifically the success rate of implementing recommendations, the proportion of accepted suggestions, and the number of sessions. The average session duration was not a significant predictor, suggesting that time alone does not reflect the effectiveness of interaction with the system. At the same time, these findings should not be interpreted as evidence that AI systems can fully replace the deeper pedagogical and creative dimensions of piano instruction. Within the framework of music pedagogy, technical accuracy is regarded primarily as a foundation for expressiveness and artistic interpretation (Jia, 2023), whereas the capacity of AI to assess interpretative depth, stylistic sensitivity, expressive nuance, and performer–audience interaction remains highly limited (Lerch et al., 2020; Liao, 2025; Wei, 2021).

These findings support Hypothesis 1 (H1): learner autonomy can be enhanced through the use of immediate AI feedback. This aligns with the theory of self-regulated learning (Zimmerman, 2002), the feedback model (Hattie and Timperley, 2007), and self-determination theory (Ryan and Deci, 2000). Hypothesis 2 (H2) was also supported, as the experimental group showed significantly greater gains in technical accuracy, consistent with the conceptual framework of human–AI partnership in education (Algammaz, 2025; Cukurova et al., 2019; Goldie, 2016). Hypothesis 3 (H3) was also confirmed: students who engaged more frequently and effectively with the AI system (more sessions, higher acceptance and implementation rates) exhibited stronger gains in both autonomy and technical accuracy, likely enhancing self-regulatory processes (Zimmerman, 2002) and activating intrinsic motivation through perceived progress. However, the observed increase in learner autonomy warrants careful interpretation, as excessive reliance on algorithmic feedback may undermine students' ability to make independent artistic decisions, potentially leading to a more standardized approach to performance preparation. While

structured feedback likely contributes to technical consistency, there remains a risk of reducing musical interpretation to a set of parameters more readily measurable by AI-based systems.

The current findings are consistent with research conducted in the Egyptian context (Algammaz, 2025), where the PFS-AI system facilitated more accurate performance evaluation through automated analysis, supporting the present study's conclusion that automated methods contribute to learning progress. Another study by Yang (2025) employing CNN models based on OpenPose showed that AI can accelerate skill development in music performance. That study reported a 33.3% increase in technical acquisition speed and an 18.6% improvement in posture over a 16-week period, suggesting that real-time, personalized feedback is a key mechanism of AI's effectiveness. The current study revealed a similar trend in regression models, indicating that the effectiveness of AI feedback is not solely determined by frequency of use but also by how students apply the feedback they receive. This parallels findings by Yang (2025), who reported a 40.2% increase in self-correction and a 95.1% retention rate for proper technique six months after system implementation. At the same time, from the perspective of performance pedagogy, excessive emphasis on technical optimization may unintentionally result in a mechanically precise yet less interpretively flexible performance style. This concern is particularly relevant in piano education, where the principal objectives of the educational process include the development of artistic individuality, mastery of phrasing, tonal imagination, and the capacity for expressive communication (Liao, 2025).

Another study by Li (2022) found that training methods based on deep convolutional neural networks and automatic note-onset detection significantly increased learner interest and engagement, with 80–85% of participants responding positively to AI-based lessons – corresponding to the autonomy gains observed in the present study (AELS delta = 6.15 in

the experimental group). Wang (2025) also reported that immediate AI feedback substantially enhanced self-efficacy and musical skills compared to traditional instruction. In that study, self-efficacy scores increased from 51.13 to 57.38 ($p = 0.012$), mirroring the current finding that the quality of system engagement (success and acceptance of recommendations) significantly predicted autonomy gains ($B = 2.85$ and $B = 2.10$, respectively).

Additionally, Chang and Peng (2022) highlighted the role of multimodal analysis and gesture classification in accurately capturing performance quality. Their proposed MSA-WCN algorithm achieved 99.12% accuracy in classifying piano gestures, which aligns with the use of MediaPipe in the current study. Another relevant study by Zhang (2024) on finger health and the risks of intensive piano training found that an AI-based system achieved 97.8% concordance with expert assessments and identified signs of fatigue and strain. This supports the notion that AI systems not only enhance technical skill development but also serve as tools for injury prevention, thereby expanding their potential applications. However, it should be noted that high classification accuracy in technical monitoring does not necessarily guarantee an equally high level of effectiveness in evaluating the expressive or interpretative characteristics of performance. Musical communication is largely characterized by subtle nuances in tempo, dynamics, and emotional delivery, which are often difficult to describe or fully standardize with algorithms, and this limitation should be carefully considered when working with AI-based feedback systems.

One study by Ou et al. (2025) examining the impact of an AI application on self-efficacy and self-regulated learning among violin students reported a substantial increase in musical performance self-efficacy in the experimental group, whereas the control group exhibited a natural decline in confidence as tasks became more complex. The present findings reflect a similar trend: autonomy increased significantly more in the

experimental group than in the control group ($\Delta = 6.15$ vs. $\Delta = 0.32$), and the success rate of implementing AI-generated recommendations emerged as a significant predictor of autonomy ($B = 2.85$, $p < 0.001$).

Another study by Lu (2025) demonstrated that AI systems integrating audio analysis, feature extraction, and predictive modeling can deliver adaptive feedback that influences expressiveness, dynamics, and performance accuracy – findings consistent with the current results. In a related investigation by Xu and Fan (2024), the focus was on novice pianists who struggled to assess their own performance quality and identify errors. The AI system used in that context was shown to provide effective feedback, enabling learners to detect mistakes, receive real-time corrections, and avoid the accumulation of incorrect motor patterns. These observations align with the present study, where technical accuracy improved more notably in the experimental group, and regression analysis revealed a significant relationship between the quality of system interaction and performance gains. In addition, another study by Tang (2025) explored a support vector machine-based AI model for analyzing learning behavior and generating personalized learning pathways. The study reported an average 10% improvement in student achievement and a reduction in error rates, reinforcing the findings of the present research.

At the same time, a study by Yeh et al. (2025) found that AI-supported instruction significantly increased learner interest but did not produce substantial differences in academic performance compared to traditional methods. This contrasts with the current study, where the performance indicators were measured objectively using Onsets & Frames and MediaPipe, providing quantitative metrics that may be more sensitive to structured, detailed feedback. In contrast, performance measures in the Yeh et al. (2025) study may have reflected higher variability or greater dependence on the creative aspects of performance, potentially explaining the lack of significant

differences. This discrepancy further demonstrates that, at the present stage of development, AI systems are more effective in supporting measurable technical domains than in addressing tasks that require interpretation and creative engagement.

Further insight can be drawn from another study by Oguguo et al. (2022), which examined the differential effects of personalized GAI-based feedback based on students' initial academic performance. The authors concluded that students with lower academic achievement benefited more from AI-generated suggestions, whereas those who were already high-performing sometimes experienced a reduction in autonomy. Conversely, fully autonomous, on-demand use of AI feedback improved outcomes for high-achieving students but reduced independence among students facing academic difficulties.

In the context of the present study, the significance of factors such as successful implementation of recommendations and the proportion of accepted feedback suggests, when compared with Oguguo et al. (2022), that the educational effectiveness of AI systems depends on the learner's level of awareness and the strategic approach they take in engaging with AI-generated feedback. In conclusion, AI-based feedback systems can serve as useful supplementary pedagogical tools; however, they cannot substitute for human guidance in the field of art. In contemporary piano pedagogy, the role of the teacher extends far beyond the mere correction of technical skills, encompassing emotional interaction, interpretative modeling, and the development of artistic individuality. At present, all of these dimensions remain difficult to replicate adequately with modern AI technologies.

4.1 Limitations

The study is limited by several factors. First, the sample consisted solely of students from a single institution, within one age group, and in a specific cultural and geographical

context. Second, no longitudinal measurements were included, which restricts the ability to assess long-term effects. Third, the measure of learner autonomy relied on a self-report instrument, which may be subject to subjective bias. Furthermore, although the AELS scale demonstrated satisfactory psychometric properties and high internal consistency, self-report-based assessments of autonomy may not always accurately reflect students' actual behavior during independent learning. It should also be acknowledged that some participant responses may have been influenced by social desirability bias.

Another important consideration relates to the limited ecological validity of the current experimental design. The study was conducted under strictly controlled laboratory conditions, which cannot fully replicate the authentic context of piano instruction in conservatory settings or home-based practice environments. Consequently, the results observed during the experiment may have been partially dependent on the available technological infrastructure, and their reproducibility in less formalized educational settings remains uncertain.

Additionally, system log data were unable to capture motivational, emotional, or personality-related factors, which may have significantly influenced the constructs under investigation. The PFS-AI system may also have limitations related to repertoire selection, the incorporation of certain performance conventions into the training data, and an emphasis on technically normative performance, all of which may have contributed to a degree of evaluative bias in the feedback generated by the system. Another limitation is the lack of comparison with alternative AI-based feedback systems, despite the existence of many such tools. Furthermore, the potential influence of instructors cannot be entirely ruled out, as the study employed a hybrid instructional model despite the

provision of standardized guidelines. Future research should incorporate more diverse and extensive samples, include retention-based measures, and adopt a multifactorial analytic approach that accounts for additional variables likely to affect the outcomes.

5. Conclusions

The study confirmed the impact of the AI-based feedback system (PFS-AI, Onsets & Frames, MediaPipe) on the development of learner autonomy and technical accuracy among piano students. The findings demonstrated that, compared with the traditional instructional approach, students in the experimental group achieved greater gains in autonomy ($\Delta = 6.15$ versus $\Delta = 0.32$ in the control group) and improved across all dimensions of technical proficiency. Mixed-design ANOVA confirmed significant main effects of time and group, as well as their interaction, while regression analysis identified key predictors of growth in autonomy and technical accuracy.

The theoretical significance of the study lies in its demonstration of the potential of student-AI feedback system partnerships to complement traditional instruction by enhancing both technical precision and learner independence. Additionally, the study contributes to the growing body of empirical and theoretical work on AI-supported music education. The practical value of the research is in its empirical validation of the effectiveness of integrating an AI feedback system into piano instruction. Such a system can be applied in online learning, individual lessons, and conventional classroom settings, offering a promising approach to reducing teacher workload.

Future research should consider expanding the participant sample, examining the long-term effects of AI-based instruction, and analyzing the influence of motivational and emotional factors. Further integration of biomechanical and physiological

data may also provide a deeper understanding of the constructs involved. Another promising direction is to explore individual differences in self-regulation skills, performance levels, and learner characteristics in relation to engagement with AI-based feedback systems.

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Research ethics committee approval

The study adhered to the ethical principles outlined in the Declaration of Helsinki. Ethical approval for the project was granted by the Ethics Committee of the Central Conservatory of Music (Beijing, China).

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